

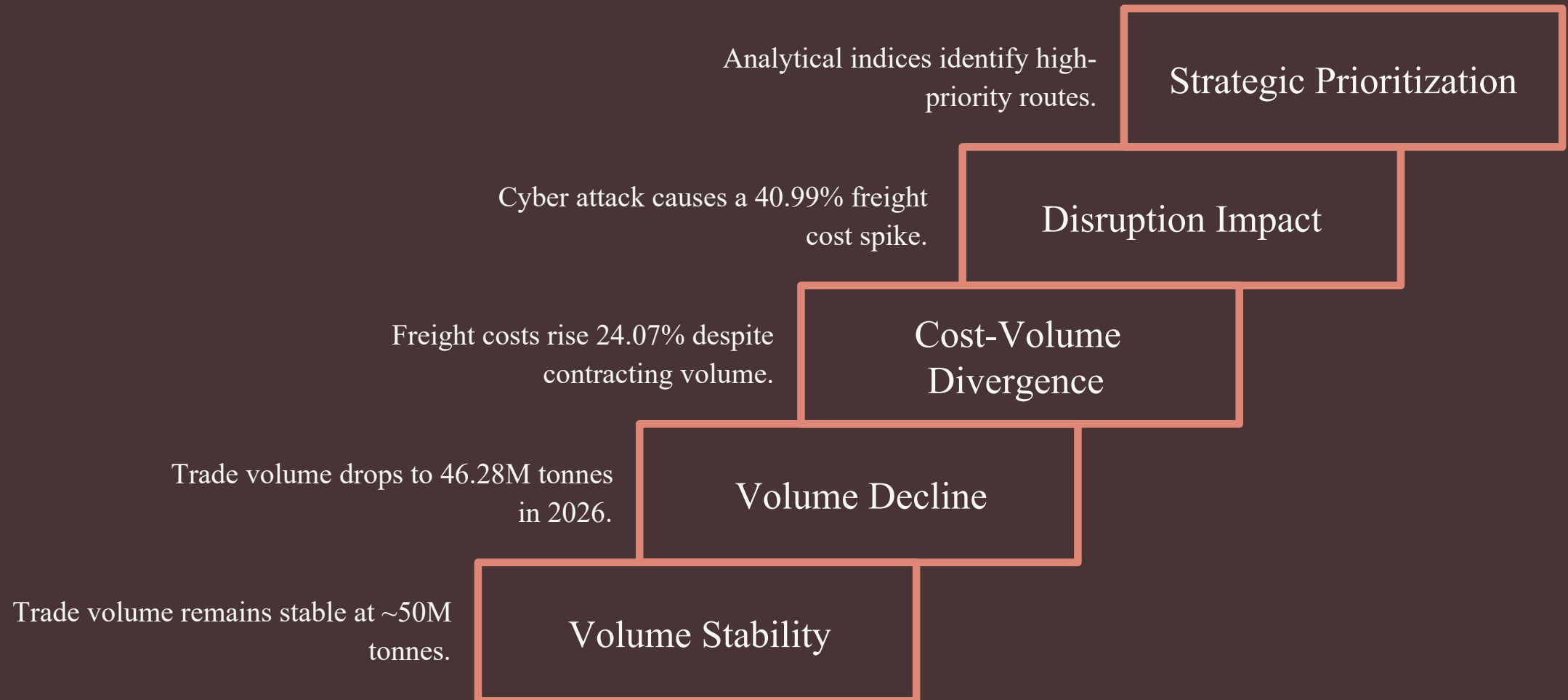
SQL Analytics
Portfolio

Global Supply Chain Disruption Analysis

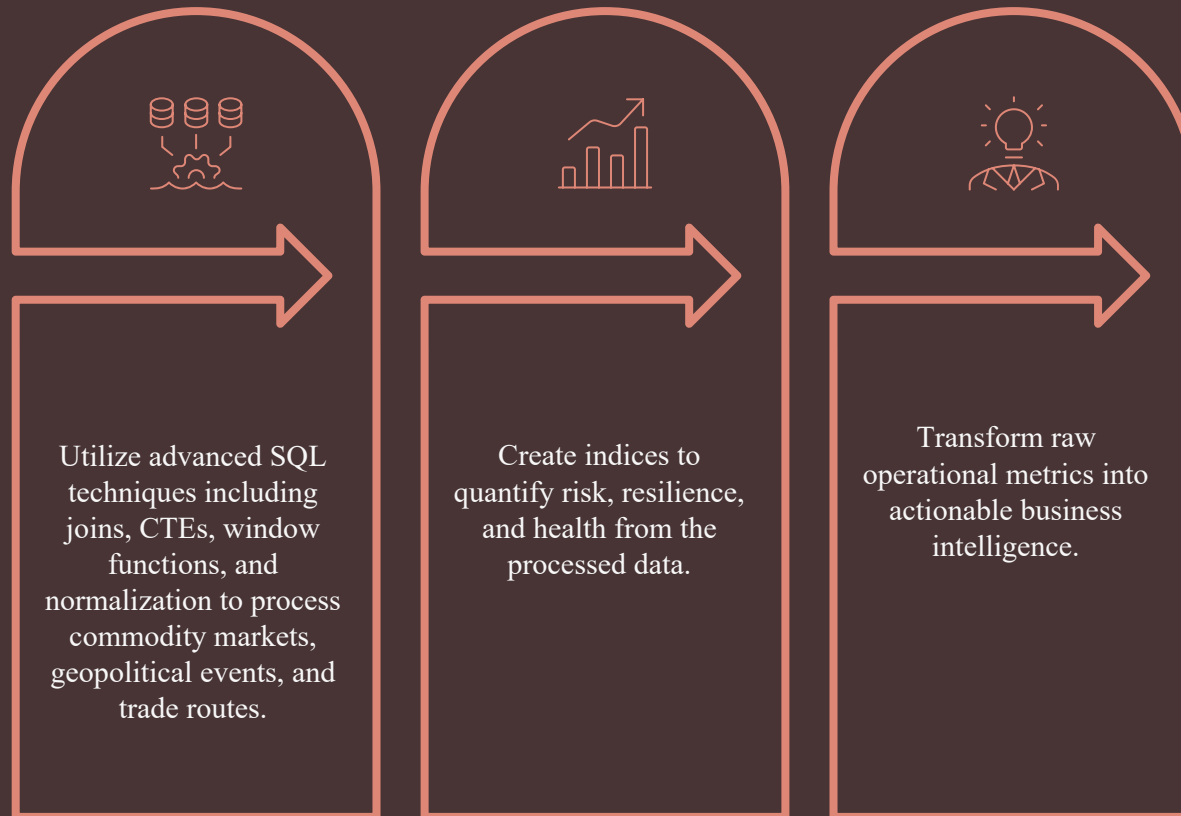
Identifying disruption
patterns, cost pressure, and
route risk | 2015–2026

SQL analytics portfolio
project | 2015–2026

Navigating Supply Chain Challenges



Leveraging SQL for Supply Chain Modeling



Trade Volume and Cost Performance



○
5.24% Trade
Volume YoY
Decrease in trade
volume

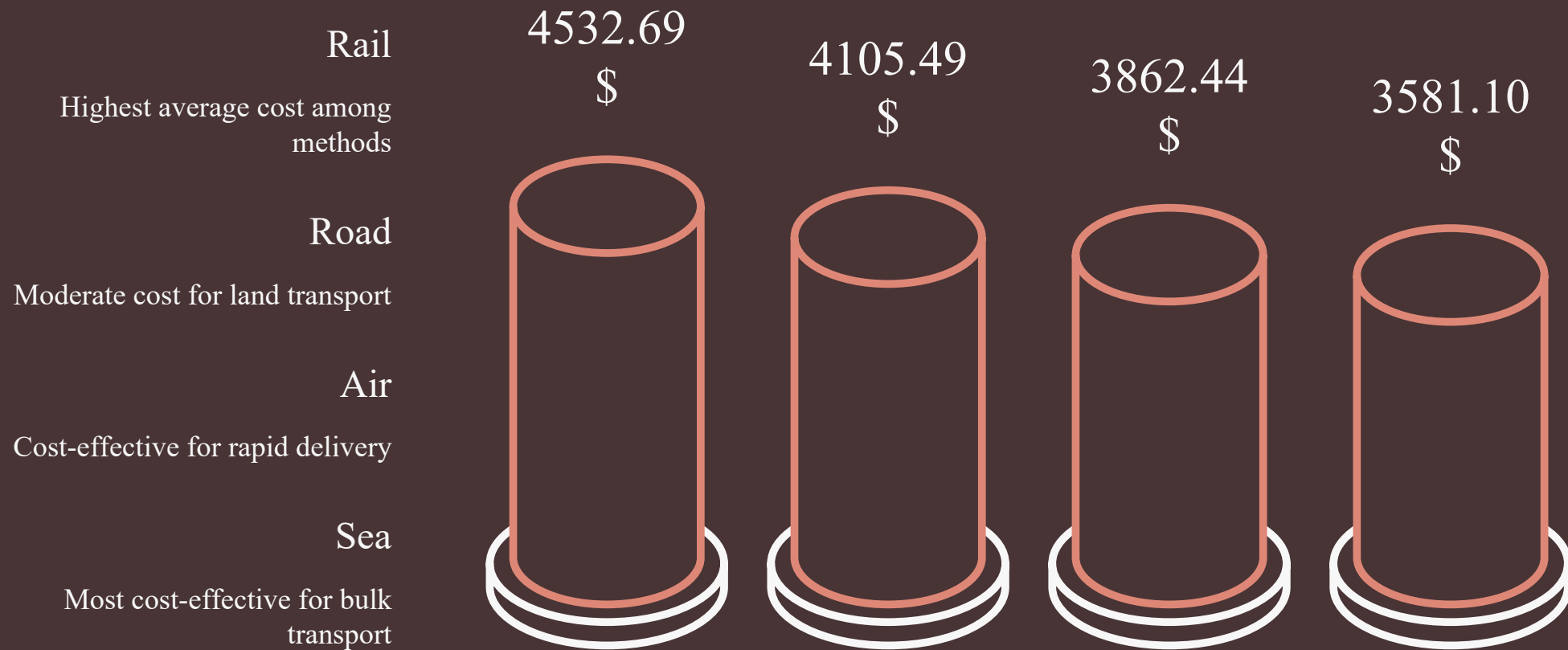
○
24.07% Freight
Cost YoY
Increase in freight cost

○
70.69% Others

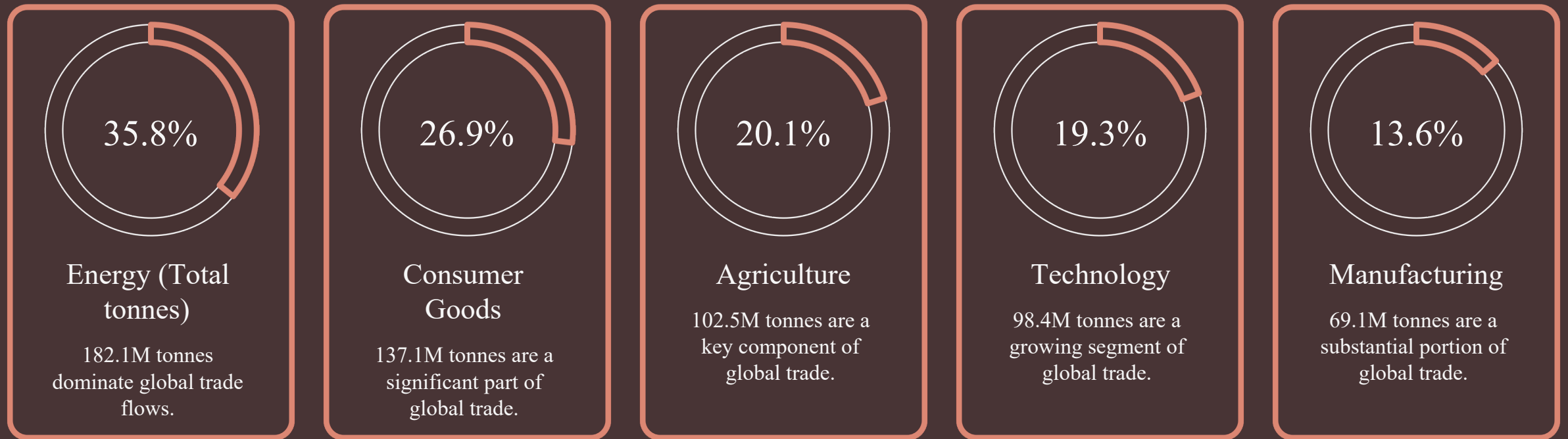
A decade of stability ended in 2026 with a sharp divergence between volume and cost

Peak trade volume reached 50.23M tonnes in 2023.

Average Freight Costs by Shipping Method



Global Trade Route Types by Volume



Energy and Consumer Goods represent the largest share of global trade volume.

Country Logistics and Dependency

India



86.66 LPI

Recorded the highest performance score in the dataset



23.73% Dependency

Maintains a lower level of reliance on external trade

France



36.64 LPI

Demonstrates a significantly lower logistical efficiency rating



98.66% Dependency

Shows the highest trade dependency among the analyzed nations

USA: 33.87 LPI | 97.90
Dependency

These metrics are critical for assessing national supply chain vulnerability and resilience.

Geopolitical Events and Commodity Stress

Regional event counts and
seasonal price pressures drive
volatility

1,730

Geopolitical events

Geopolitical events in
Europe (Highest regional
count)

53.08

Commodity stress
index

Peak commodity stress
index (June)

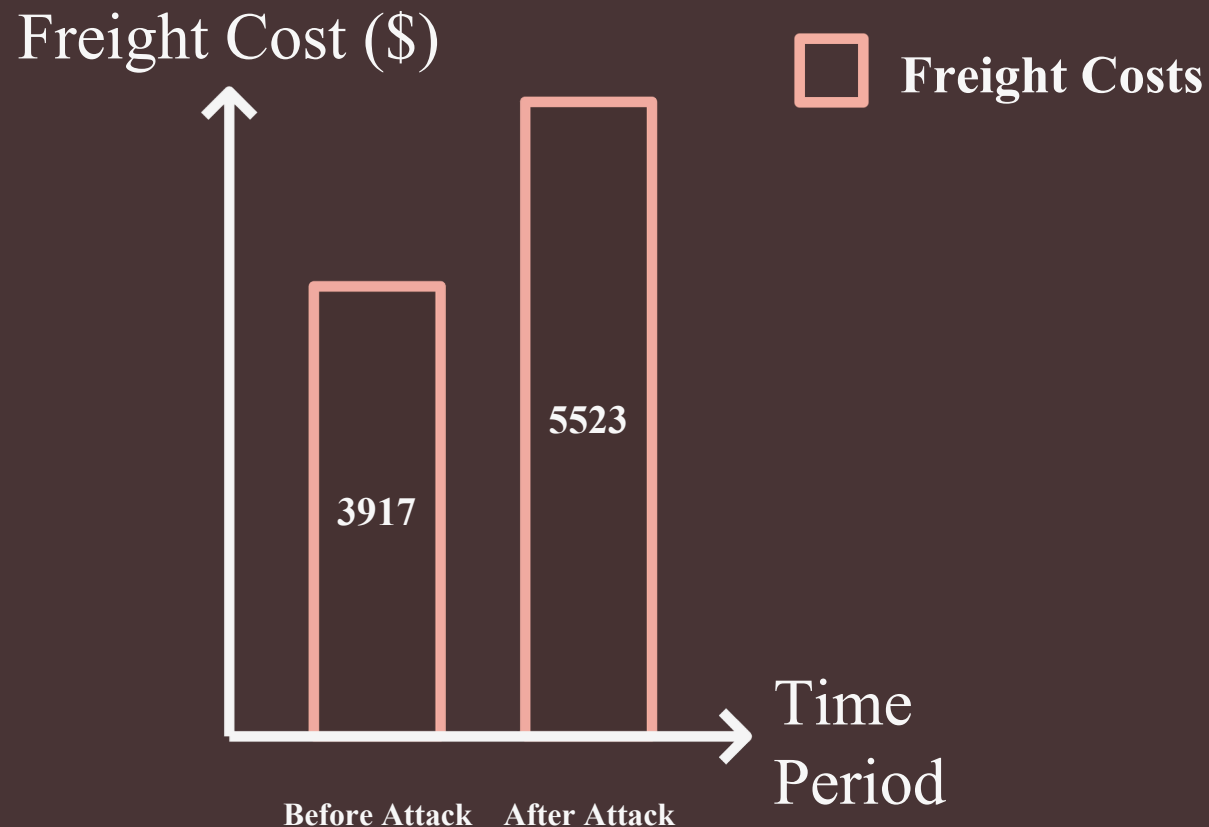
87.46

Oil price pressure

Average Oil price pressure
in 2022 and natural gas
at 6.85 on the annual
averages supplied.

Impact of Major Geopolitical Events

A Cyber Attack on 2026-06-04 resulted in a 40.99% increase in freight costs within a 28-day window.



Port Congestion and Shipping Delay

Congestion Level	High Congestion	Medium Congestion	Low Congestion
Average Delay	8 days	6 days	5 days

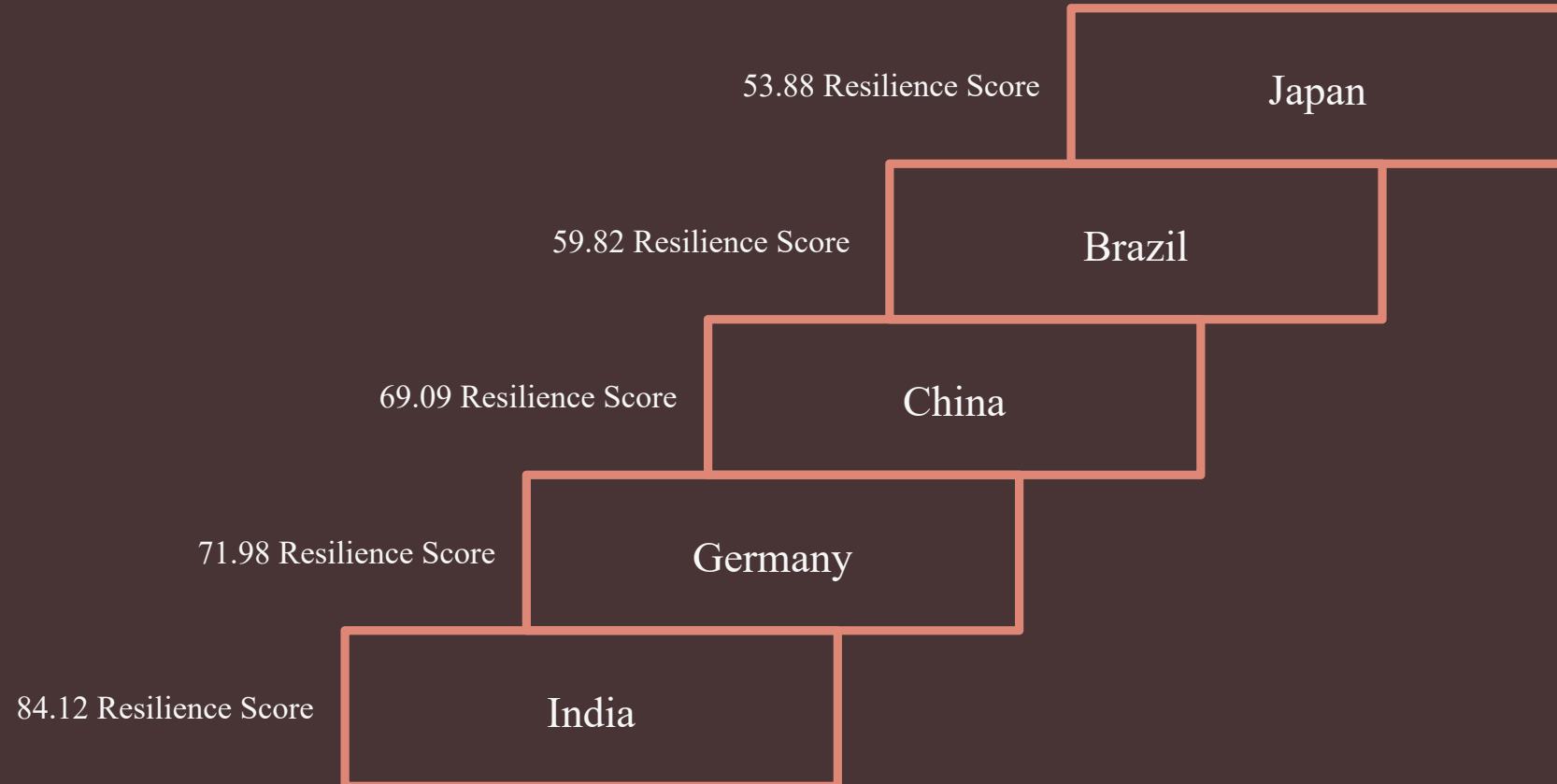
Top High-Risk Trade Routes

Ranking routes by risk score, congestion, and operational delays

1	Assess Route R00043 Route R00043 holds a 66.25 Risk Score, ranking 1st.
2	Assess Route R00037 Route R00037 holds a 65.87 Risk Score.
3	Assess Route R00044 Route R00044 holds a 65.87 Risk Score.
4	Assess Route R00020 Route R00020 holds a 65.67 Risk Score.
5	Assess Route R00032 Route R00032 holds a 65.66 Risk Score.

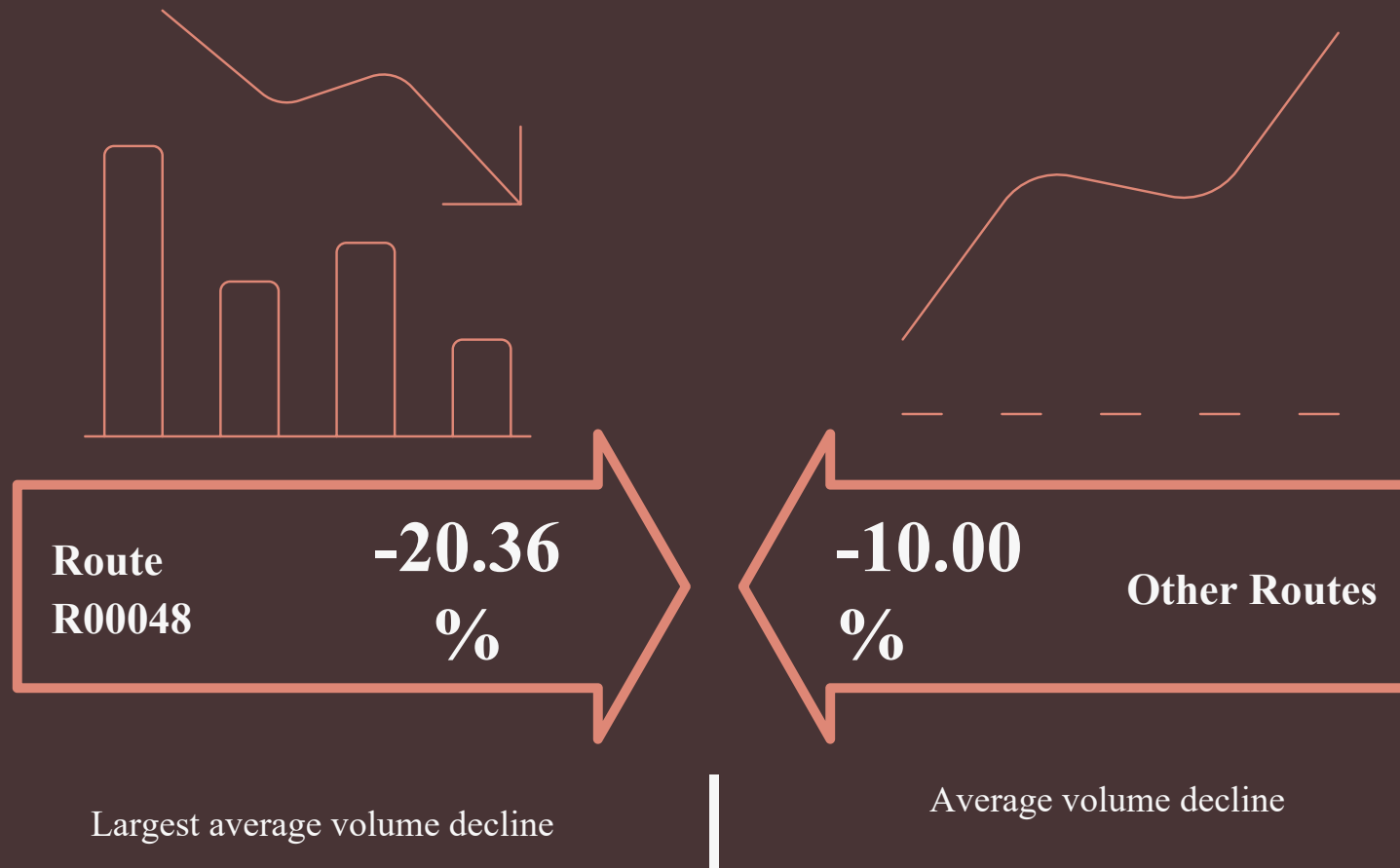
Country Supply Chain Resilience

Ranking nations by LPI, port capacity, and trade dependency

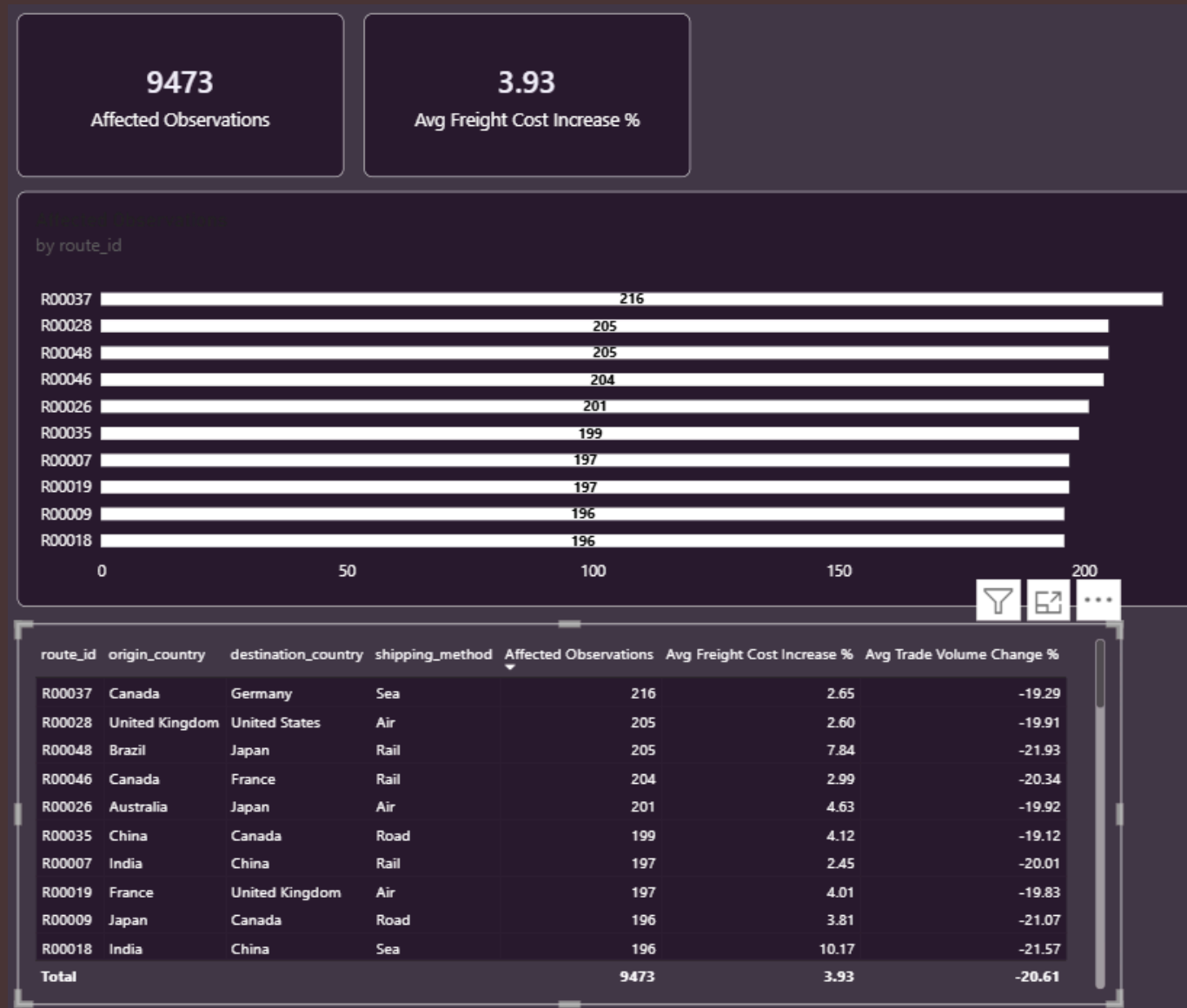


Commodity Stress vs. Volume Decline

Over 7,800 observations
showed concurrent high stress
and declining volume.



Freight cost vs Declining trade volume



Moving Average and Cumulative Trade Volume



2016

Total trade volume in 2016

2021

Total trade volume in 2021

2026

Total trade volume in 2026

4-Week Moving Average —
Used to detect sustained
freight-cost increases.

Analysis of long-term trends
using moving averages and cu-
mulative volume to smooth out
weekly volatility.

Operational Continuity

281

Operational Weeks

281 Operational Weeks (Route R00028)

1,967

Operational Days

1,967 Operational Days (Uninterrupted)

Stability

Benchmark

Benchmark for stability in the global network.

Measuring uninterrupted route activity as a benchmark for stability

Geopolitical Exposure by Country

France, the US, and Australia lead in exposure scores due to high trade dependency and event severity.

country	region	trade_dependency_score	avg_event_severity	geopolitical_exposure_score
France	Europe & Central Asia	98.66	5.49	541.66
United States	North America	97.90	5.52	540.39
Australia	East Asia & Pacific	88.80	5.57	494.19
Brazil	Latin America & Caribbean	69.47	5.41	375.53
Canada	North America	57.34	5.52	316.51
Total		412.17	27.51	2,268.28

01

France

541.66 Score | 98.66% Dependency

02

United States

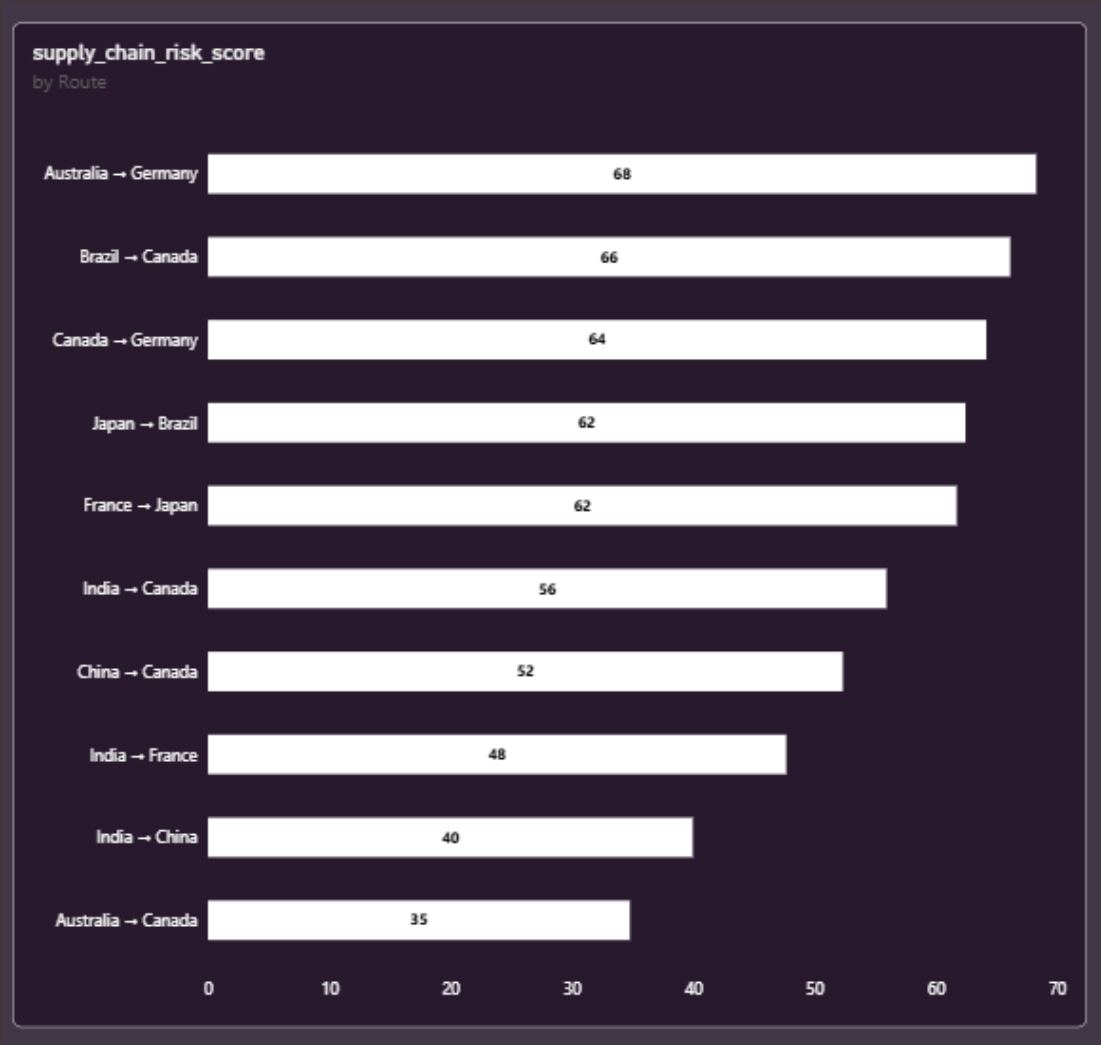
540.39 Score | 97.90% Dependency

03

Australia

494.19 Score | 88.80% Dependency

Supply Chain Risk Score



Route Resilience Score

route_id	origin_country	destination_country	shipping_method	trade_route_type	resilience_score
R00013	Australia	Canada	Sea	Energy	78.58
R00009	Japan	Canada	Road	Energy	75.64
R00028	United Kingdom	United States	Air	Technology	72.33
R00016	United States	France	Rail	Energy	70.04
R00049	United Kingdom	China	Air	Consumer Goods	66.63
R00023	United States	Japan	Air	Consumer Goods	64.90
R00015	United Kingdom	Australia	Road	Consumer Goods	61.83
R00034	France	United States	Rail	Manufacturing	61.34
R00010	Germany	India	Rail	Energy	60.99
R00041	United States	Japan	Road	Energy	60.93
Total					673.21

Supply Chain Health Index

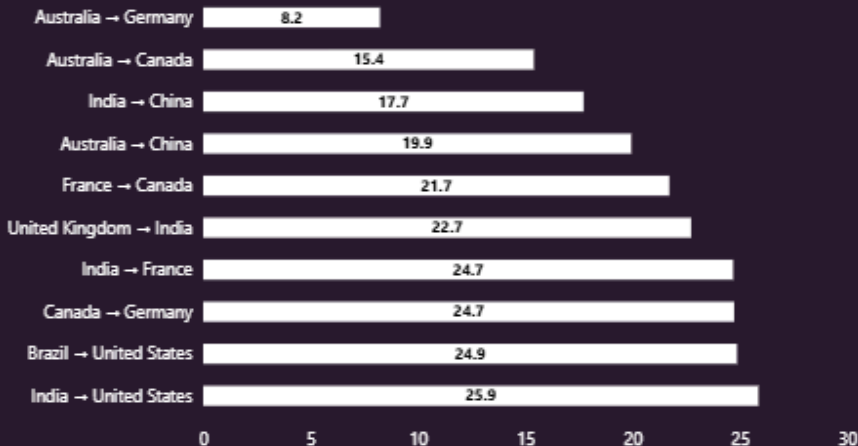
Top 10 Healthiest Routes

health_index by Route



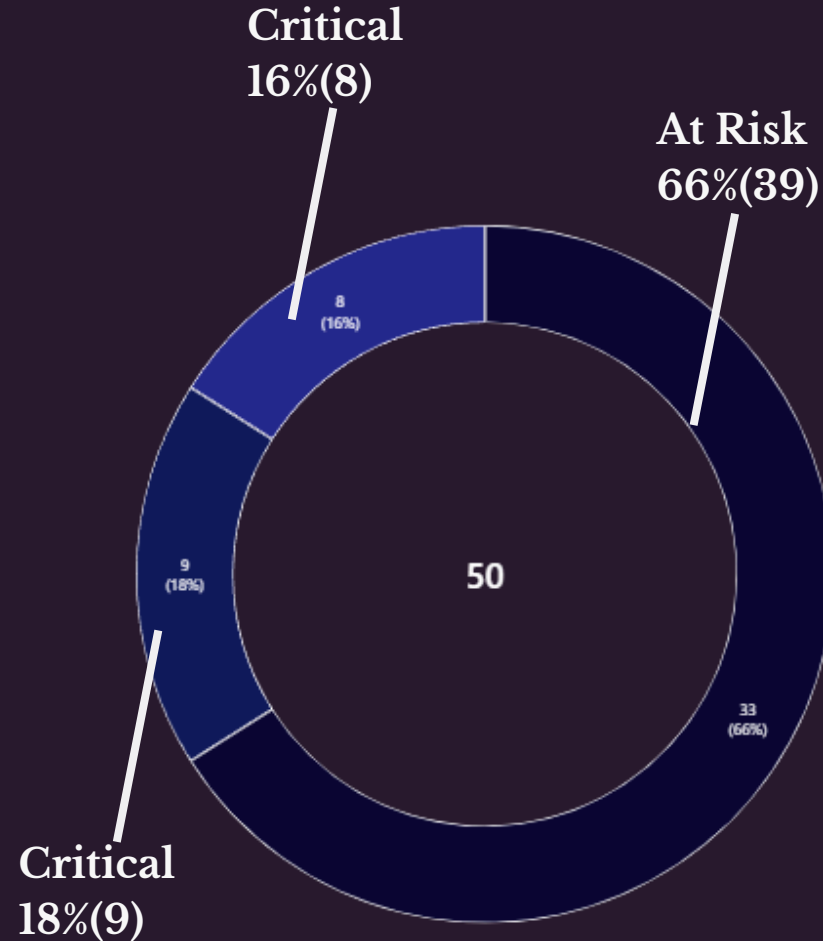
Bottom 10 Most Vulnerable Routes

health_index by Route

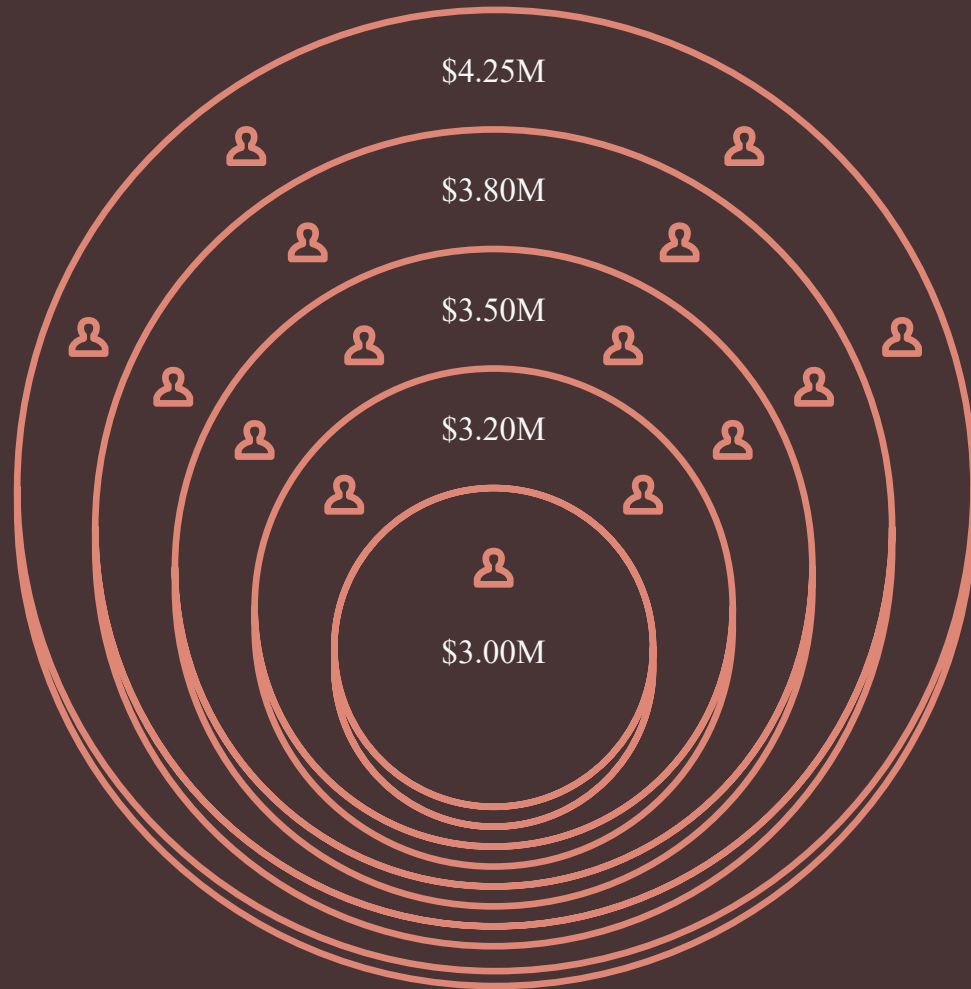


Health Status Distribution

route_id by health_status



Freight Cost-Based Route Value



UK → USA (Sea,
Energy)

Top route by freight cost

Route R00008

Second highest route value

Route R00024

Third highest route value

Route R00043

Fourth highest route value

Route R00041

Fifth highest route value

Note: This represents cost-based
route value, not net profit.

Management Priorities and Strategy

Investigate 2026 Divergence

Investigate 2026 cost pressure: freight cost +24.07% while trade volume -5.24%.

Prioritize High-Risk Routes

Prioritize high-risk routes and identify resilient alternatives.

Early Warning Systems

Monitor 4-week moving averages to detect sustained cost increases.

Contingency Planning

Review countries with high trade dependency and high geopolitical exposure for diversification or contingency planning

Methodology and Data Quality Notes



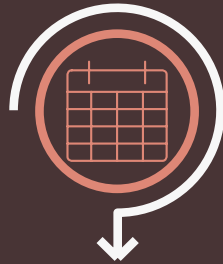
Custom Models

Weights and normalization rules are dataset-specific.



Metric Ranges

LPI and Port Congestion indices are internal benchmarks.



Temporal Analysis

Weekly comparisons use LAG functions (WoW), not YoY.



Causality

Correlation or association does not prove causation.

Methodology Details

Composite risk, resilience, and health scores are custom analytical models.

Final Business Story

The supply chain remained broadly stable in volume for many years, but 2026 shows a clear deterioration in cost and volume performance.

Geopolitical and commodity pressures create uneven exposure across countries and routes.

Some routes repeatedly experience high commodity stress while volumes decline, creating candidates for intervention.

Risk, resilience and health scores convert many operational metrics into decision-ready route priorities.

The next step is to connect these findings to actions: route diversification, supplier alternatives, inventory buffers, carrier strategy and early-warning monitoring.
